Alex Bondarenko

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EXPERIENCE

Tempus Labs | Bioinformatics Software Engineer 2021 — Present

- Maintained and built dozens of containerized bioinformatics pipeline components, Python libraries, CLI tooling, internal services, and data pipelines processing large-scale genomic datasets which have been deployed into IVD and LDT medical devices that handle 30k patient samples/week
- Collaborated with clinical scientists to develop algorithms and rules engines for variant pathogenicity prediction and interpretation, allowing automatic bypass of new variants in the thousands and decreasing weekly review burden by hours each week
- Protyped an LLM agent to interact with the MSK Oncology knowledge base using SQL querying and OpenAPI tools with Langchain, later deployed to an internal AI portal
- **Key skills**: Data Engineering (Pandas, Dask, Dagster, Pydantic), FastAPI, CI/CD (Docker, Jenkins, Terraform, GCP Concourse), Bioinformatics (reproducible research with Nextflow, Snakemake)

Google | Software Engineering Intern Apr 2018 — Jul 2018

- Worked with the Device Inventory team on numerous data engineering and information retrieval tasks for a schemaless database of corporate device assets
- Designed and implemented autocomplete for an internal device search engine written in Django, impacting 2,000 monthly active users across finance, security, and support
- Architected indexing pipelines using MapReduce and key-value stores

Intuit | Software Engineering Intern Apr 2017 — Jul 2017

- Fullstack work that powered performance analytics for core infrastructure
- Engineered a diagnostic tool that monitors Java Spring Boot applications in deployment to streamline the profiling of real-time applications
- Designed an easy-to-use single-page dashboard powered by jQuery that interacts with custom API endpoints that expose metrics and control logging
- Spearheaded the evaluation and integration of a distributed tracing solution to unify logging across business units

EDUCATION

UC Berkeley 2016 — 2020

Bachelor - Computer Science

- Machine Learning, Genomics, Computational Neuroscience, Network Science (Python)
- Embedded Systems (C)
- UPE (CS Honor Society)
- High Performance Computing (OpenMP/CUDA C++)
- Database Systems, Data Structures, Synthetic Biology (Java)
- Time Series Analysis, Econometrics (R)
- Security Engineering (Go)

PROJECTS

SVM: Contrastive learning for multi-modal protein embeddings May 2023 — May 2023

- Worked with an international team to create a unified protein embedding by linking multiple modalities (AA and NT sequences, 3D structure, GO terms) with the goal of improving downstream tasks like pathogenic mutation prediction
- Won 1st place and impact prizes out of 600+ attendees and 17 final projects ranging from LLMs to explore genomics to foundation models for single cell biology
- Key contribution was building out the data ingestion, training, and inference pipeline executed via Runpod GPU cluster, and prototyping a UI with Atlas and Gradio which I deployed to Huggingface Spaces via Github Actions

Data Science Competitions | Kaggle

- Placed in top 15% in a competition to predict thermostability of single-point protein mutations. Used ESMFold and XGBoost primarily.
- Placed in top 100 in a CTF to perform adversarial attacks against machine learning models.